

Yang–Mills Vacuum Stability as a Classical Spectral Property

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Abstract

We present a formal verification that the stability of the Yang–Mills vacuum — the existence of a positive mass gap — follows from classical Sturm–Liouville spectral theory on a weighted Hilbert space $L^2((0, \infty), \omega)$. The proof chain consists of five Z3-verified lemmas: (A) the anomalous dimension $\gamma > 0$ places the ITCM weight in the confining class $\beta > 1$; (B) the Hankel asymptotic correspondence gives confining potential growth $V(x) \sim x^{2\gamma} \rightarrow +\infty$; (C) Weyl’s limit-point criterion (1910) ensures essential self-adjointness; (D) the Rellich–Molchanov theorem guarantees purely discrete spectrum; (E) discrete spectrum with ordered eigenvalues gives a positive spectral gap $\Delta > 0$. The full chain is consolidated into a single Z3-verified theorem: $\gamma > 0 \Rightarrow \Delta > 0$. For the physically relevant case $\gamma = 1/2$ (linear confinement), the gap scales as $\Delta = 1.7498 \cdot \sigma^{2/3}$ from the Airy zeros. The entire mechanism uses mathematics no later than 1953 (Molchanov). No path integrals, no Born rule, no infinite-dimensional measure theory appears in the proof. The single physics input is the sign of the gluon anomalous dimension, supported by perturbative QCD [12], lattice simulations [13], and Dyson–Schwinger equations [14]. This does not solve the Clay Millennium Problem — Assumption E (the existence of a rigorous 4D Yang–Mills theory whose radial ITCM sector matches the scaffold) remains open. The contribution is the identification of the mass gap mechanism as a classical geometric necessity: a second-order ODE on a half-line with a growing restoring force. All 34 examples across 12 structures are verified by the Z3 SMT solver in the Kleis formal verification language.

Keywords: Yang–Mills mass gap, vacuum stability, Sturm–Liouville, spectral gap, classical spectral theory, Weyl limit-point, Rellich–Molchanov, Airy function, formal verification, Z3

1 Introduction

The Yang–Mills mass gap problem asks whether four-dimensional quantum Yang–Mills theory with gauge group $SU(N)$, $N \geq 2$, has a positive mass gap $m > 0$ in the spectrum of its Hamiltonian [1]. It is one of the seven Clay Millennium Prize Problems, and no rigorous proof exists despite decades of progress in perturbative QCD, lattice simulations, and non-perturbative methods.

This paper advances a specific claim: the *mechanism* that produces the mass gap is classical. It belongs to Sturm–Liouville spectral theory on a half-line, uses tools from the first half of the 20th century, and requires no quantum field theory apparatus.

The claim rests on the Integral Transform Composition Method (ITCM) framework developed in Volumes VII–VIII of the Projected Ontology Theory (POT) series [2, 3]. The ITCM constructs a transmutation operator $T_w = H_\nu^{-1} \circ M_w \circ H_\mu$ from Hankel transforms H_μ, H_ν and a multiplication operator M_w . When the weight function $w(k)$ has an infrared singularity $w(k) \sim k^{-2(1+\gamma)}$ with $\gamma > 0$, the resulting kernel is the Green’s function of a Sturm–Liouville operator $L = -d^2/dx^2 + V(x)$ with confining potential $V(x) \rightarrow +\infty$ [3].

The single physics input is:

$$\gamma > 0 \quad (\text{sign of the gluon anomalous dimension}).$$

Everything else is mathematics. The proof chain is:

$$\gamma > 0 \Rightarrow \beta > 1 \Rightarrow V(x) \rightarrow +\infty \Rightarrow \text{self-adjoint} + \text{discrete spectrum} \Rightarrow \Delta > 0.$$

Each step is a Z3-verified [15] algebraic implication. The classical theorems (Weyl [9], Rellich–Molchanov [7, 10]) enter as labeled geometric hypotheses, and Z3 verifies the algebraic consequences. The full chain is consolidated into a single verified theorem with 34 examples across 12 structures.

2 The Bessel–Sturm–Liouville Operator

The operator central to this work is the Bessel–Sturm–Liouville operator on the half-line $(0, \infty)$:

$$L_\nu = -\frac{d^2}{dx^2} + \frac{\nu^2 - 1/4}{x^2} + V_{\text{conf}}(x), \quad u(0) = 0,$$

where $\nu \geq 0$ is the Hankel order and $V_{\text{conf}}(x)$ is the confining potential.

The centrifugal barrier $(\nu^2 - 1/4)/x^2$ is the classical angular momentum term. For $\nu > 1/2$, it is strictly positive and prevents the eigenfunction from collapsing to the origin (Z3-verified: `CentrifugalBarrier` structure).

The confining potential $V_{\text{conf}}(x) \sim \sigma \cdot x^{2\gamma}$ grows without bound for $\gamma > 0$, preventing the eigenfunction from escaping to infinity. Together, the two terms *trap* the particle — creating the gap.

In the ITCM dictionary, the Hankel-order asymmetry $\mu \neq \nu$ corresponds to the non-abelian gauge coupling of $\text{SU}(N)$. The abelian case $\mu = \nu$ reduces to the free Bessel operator with no confinement (Z3-verified: `HankelOrderAsymmetry` structure). The non-abelian case introduces the anomalous dimension $\gamma > 0$ that drives confinement.

$$L = -\frac{d^2}{dx^2} + \underbrace{\frac{\nu^2 - 1/4}{x^2}}_{\text{centrifugal}} + \underbrace{\sigma \cdot x^{2\gamma}}_{\text{confining}}$$

3 The IR Regularity Chain

The proof consists of five lemmas, each verified by the Z3 SMT solver. We present the chain in logical order.

3.1 Lemma A: IR Exponent Classification

The ITCM weight function has infrared behavior $w(k) \sim k^{-2\beta}$ as $k \rightarrow 0$, where $\beta = 1 + \gamma$ is the IR exponent. The anomalous dimension $\gamma > 0$ gives $\beta > 1$, placing the weight in the *confining class*.

Z3 verification: IRExponentClassification structure. Axioms: $\gamma > 0$, $\beta = 1 + \gamma$. Verified: $\beta > 1$.

This is purely algebraic. The physics enters only through $\gamma > 0$.

3.2 Lemma B: Confining Potential Growth

The Hankel asymptotic correspondence (Watson [6], Sitnik–Shishkina [5]) maps infrared singularities to position-space growth:

$$w(k) \sim k^{-2\beta} \Rightarrow V(x) \sim x^p, \quad p = 2(\beta - 1).$$

When $\beta > 1$: $p > 0$, so $V(x) \rightarrow +\infty$ as $x \rightarrow \infty$. The potential is confining.

Z3 verification: ConfiningPotentialGrowth structure. Axioms: $\beta > 1$, $p = 2\beta - 2$. Verified: $p > 0$.

For the Yang–Mills case $\gamma = 0.5$: $\beta = 1.5$, $p = 1$ (linear confinement, Z3-verified).

The growth law is an asymptotic class correspondence (Level B), not an exact pointwise identity. It holds under the regularity conditions established by Watson’s lemma in the companion paper [3].

3.3 Lemma C: Weyl Limit-Point Criterion (Self-Adjointness)

Classical Theorem (Weyl [9]; Reed–Simon [7], Thm. X.8). If $V(x) \rightarrow +\infty$ as $x \rightarrow \infty$, then the Sturm–Liouville operator L is in the *limit-point case* at infinity and is *essentially self-adjoint* on its minimal domain.

Physical meaning: the vacuum state is unique. There is exactly one self-adjoint extension — no boundary condition at infinity is needed. The vacuum is a mathematically stable ground state.

Z3 verification: WeylLimitPoint structure. Geometric hypothesis: $p > 0 \Rightarrow \text{self_adjoint} = 1$. Z3 confirms the flag propagates correctly through the chain.

3.4 Lemma D: Rellich–Molchanov Discreteness

Classical Theorem (Rellich [7, Ch. XIII]; Molchanov [10]). If $V(x) \rightarrow +\infty$ as $x \rightarrow \infty$, then L has *compact resolvent* and therefore *purely discrete spectrum*: eigenvalues $\lambda_0 < \lambda_1 < \lambda_2 < \dots$ accumulating only at $+\infty$.

Z3 verification: RellichMolchanovDiscreteness structure. Geometric hypothesis: confining potential. Verified: $E_1 > E_0 > 0$, $\text{gap} = E_1 - E_0 > 0$.

3.5 Lemma E: Spectral Gap Positivity

If the spectrum is discrete with $\lambda_0 < \lambda_1$, then $\Delta = \lambda_1 - \lambda_0 > 0$.

This is trivial, but it is where the statement becomes precise: the mass gap is an *eigenvalue gap* of a second-order ODE on a half-line.

Z3 verification: SpectralGapPositivity structure. Verified: $\Delta = \lambda_1 - \lambda_0 > 0$.

4 The Vacuum Stability Theorem

The five lemmas chain into a single result.

Theorem (Vacuum Stability — Classical Mechanism). *If $\gamma > 0$, then the Bessel–Sturm–Liouville operator L on $(0, \infty)$ is essentially self-adjoint with purely discrete spectrum and positive spectral gap $\Delta > 0$.*

Z3 verification: VacuumStabilityTheorem structure consolidates all five lemmas. The full chain is verified:

$$\gamma > 0 \Rightarrow \beta > 1 \Rightarrow p > 0 \Rightarrow \text{self-adjoint} + \text{discrete} \Rightarrow \Delta > 0.$$

An additional Z3-verified identity: $p = 2\gamma$. The potential growth exponent equals twice the anomalous dimension.

$$\gamma > 0 \xrightarrow{A} \beta > 1 \xrightarrow{B} p > 0 \xrightarrow{C+D} \text{self-adjoint} + \text{discrete} \xrightarrow{E} \Delta > 0$$

4.1 Quantitative Corollary: Airy Scaling

For the physically relevant case $\gamma = 1/2$ (linear confinement), $V(x) = \sigma \cdot x$, the eigenvalues are given by the Airy zeros:

$$E_n = \sigma^{2/3} \cdot |a_n|,$$

where a_n are the zeros of the Airy function $\text{Ai}(x)$. With $|a_1| = 2.3381$ and $|a_2| = 4.0879$:

$$\Delta = (4.0879 - 2.3381) \cdot \sigma^{2/3} = 1.7498 \cdot \sigma^{2/3}.$$

Z3 verification: AiryScalingCorollary structure. Verified: $1.74 < \Delta < 1.76$ at unit string tension.

The harmonic benchmark ($\gamma = 1$, $V = \omega^2 x^2$) provides an exact cross-check: $\Delta = 4\omega$, independent of the Hankel order μ (Z3-verified: HarmonicBenchmark structure).

5 The Centrifugal Barrier Mechanism

The centrifugal term $(\nu^2 - 1/4)/x^2$ is strictly positive for $\nu > 1/2$ (Z3-verified). Combined with the confining term $V_{\text{conf}}(x) \rightarrow +\infty$, the total potential traps the particle from both sides:

- The barrier prevents collapse to $x = 0$.
- The confining growth prevents escape to $x = \infty$.

In the ITCM dictionary:

QFT language	Classical SL language
Gauge group $SU(N)$	Hankel-order asymmetry $\mu \neq \nu$
Anomalous dimension γ	IR singularity exponent
Path integral	Resolvent $(L - zI)^{-1}$
Vacuum state	Ground eigenfunction φ_0
Mass gap	Spectral gap $\Delta = E_1 - E_0$
Confinement	$V(x) \rightarrow +\infty$
Non-perturbative	Exact eigenvalue problem

The single physics input $\gamma > 0$ — supported by three independent lines of evidence: perturbative QCD [12], lattice simulations [13], and Dyson–Schwinger equations [14] — is all that the classical mechanism requires.

6 Z3 Verification Summary

The complete verification comprises 12 structures and 34 Z3-checked examples:

Structure	Examples	What it verifies
IRExponentClassification	2	$\gamma > 0 \Rightarrow \beta > 1$
ConfiningPotentialGrowth	3	$\beta > 1 \Rightarrow p > 0$
WeylLimitPoint	1	$p > 0 \Rightarrow$ self-adjoint
RellichMolchanovDiscreteness	2	$p > 0 \Rightarrow$ discrete spectrum
SpectralGapPositivity	2	discrete $\Rightarrow \Delta > 0$
VacuumStabilityTheorem	3	Full chain: $\gamma > 0 \Rightarrow \Delta > 0$
AiryScalingCorollary	3	$\Delta = 1.7498 \cdot \sigma^{2/3}$
CentrifugalBarrier	2	$\nu > 1/2 \Rightarrow$ barrier > 0
HankelOrderAsymmetry	2	$\mu = \nu \Rightarrow$ abelian (no confinement)
CombinedPotential	2	Total potential exceeds each term
ITCMDictionary	2	$SU(N) \Rightarrow \mu > \nu; \gamma > 0 \Rightarrow \Delta > 0$
NumericalGapCheck	4	$\gamma \in \{0.3, 0.5, 0.7\}$: all confining
HarmonicBenchmark	3	$\Delta = 4\omega$ (exact, independent of μ)
HonestScope	2	Classical tools; no QFT; Clay not solved

All structures use the $\mathbb{R}$ (Real) sort in Z3’s [15] nonlinear arithmetic theory (QF_NRA). Each axiom is either an algebraic identity or a labeled geometric hypothesis (classical theorem). Z3 verifies the algebraic consequences; the classical theorems provide the mathematical content. For a modern treatment of the Sturm–Liouville framework see Zettl [16].

7 Discussion

7.1 What is proved and what is not

Proved (Z3-verified, Levels A/B):

1. $\gamma > 0 \Rightarrow \beta > 1$ (algebraic, Lemma A).
2. $\beta > 1 \Rightarrow p > 0$ (algebraic, Lemma B).
3. $p > 0 \Rightarrow$ self-adjoint (Weyl 1910, Lemma C).
4. $p > 0 \Rightarrow$ discrete spectrum (Rellich–Molchanov, Lemma D).
5. Discrete $\Rightarrow \Delta > 0$ (trivial, Lemma E).
6. Full chain: $\gamma > 0 \Rightarrow \Delta > 0$ (Theorem).
7. Quantitative: $\Delta = 1.7498 \cdot \sigma^{2/3}$ (Corollary, Airy zeros).
8. Centrifugal: $\nu > 1/2 \Rightarrow$ barrier > 0 (algebraic).
9. Harmonic benchmark: $\Delta = 4\omega$ (exact, Level A).

Assumed (not proved here):

1. **Assumption A:** $\gamma > 0$ (physics input from perturbative QCD [12], lattice QCD [13], and Dyson–Schwinger equations [14]). Three independent lines of evidence support $\gamma > 0$, with floor $\gamma_{\text{floor}} \approx 0.1$ after subtracting systematics [2].
2. **Assumption B (partial):** The $w(k) \rightarrow V(x)$ correspondence is an asymptotic class mapping (Level B), not an exact pointwise identity.
3. **Assumption E:** The 4D Yang–Mills theory exists and its radial ITCM sector coincides with L (Level C/D). This is the constructive QFT gap — the hardest open problem in mathematical physics.

The Clay Millennium Problem is not solved. Assumption E contains the unresolved 4D existence problem. What we have established is the *mechanism*: if the 4D theory matches the scaffold, the gap is a classical necessity.

7.2 Why the mechanism is classical

The classical tools used in the proof chain are:

1. Watson’s lemma [6] (1918) — IR-to-position asymptotics.
2. Weyl’s limit-point criterion [9] (1910) — essential self-adjointness.
3. Rellich’s compactness [7] (1948) — compact resolvent from confining V .
4. Molchanov’s criterion [10] (1953) — $V \rightarrow +\infty \iff$ discrete spectrum.
5. Titchmarsh eigenfunction expansion [8] (1946) — spectral decomposition on half-line.
6. Borg uniqueness [11] (1946) — rigidity of inverse spectral map.

The most recent result is from 1953. No path integrals, no Born rule, no infinite-dimensional measure theory, no renormalization group appears in the mechanism. The operator L is a vibrating string with a growing restoring force — the same class studied by Sturm (1836) and Liouville (1837).

The difficulty of the mass gap was never the mechanism. The difficulty is the *realization*: whether quantum Yang–Mills theory implements this classical mechanism through its radial ITCM sector.

8 Conclusion

We have verified, for the first time, that the stability of the Yang–Mills vacuum is a classical property of a weighted Hilbert space. The verification consists of 34 Z3-checked examples across 12 formal structures, establishing a complete chain from the single physics input $\gamma > 0$ to the conclusion $\Delta > 0$.

The proof chain uses only classical Sturm–Liouville spectral theory. The mass gap is the spectral gap of a second-order ODE on a half-line with a growing restoring force. The centrifugal barrier from the Hankel-order asymmetry $\mu \neq \nu$ (non-abelian gauge coupling) and the confining potential from $\gamma > 0$ (anomalous dimension) together trap the eigenfunction, creating the gap.

The contribution is not a solution to the Clay Millennium Problem — Assumption E (the 4D QFT construction) remains open. The contribution is the precise identification of the mass gap *mechanism* as a classical geometric necessity, and its formal verification through machine-checked proofs.

The tools needed to understand this mechanism have been available since 1953. The Projected Ontology framework [2, 3] and the Kleis formal verification language [4], powered by the Z3 SMT solver [15], provided the lens to see how to apply them.

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